GPUs (architecture & CUDA)

October 13, 2026

VSS Lab

CPU vs GPU?

Performance angle…

CPU >> GPU

?

GPU >> CPU

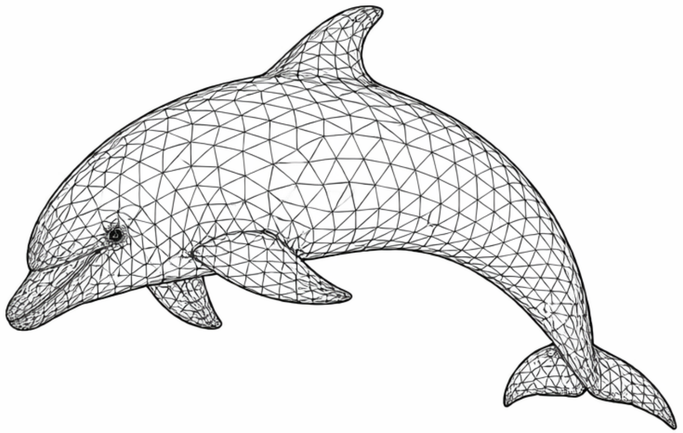
Performance angle…

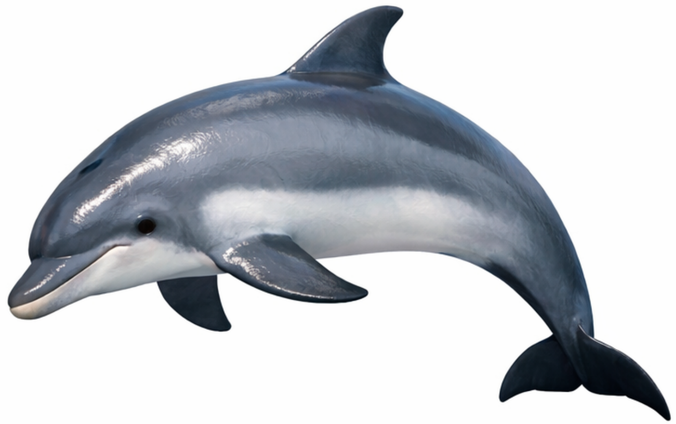
CPU >> GPU **For sequential problems**

?

GPU >> CPU **For (embarrassingly) parallel problems**

GPU Application — 3D rendering





https://gfxcourses.stanford.edu/cs248a/winter26

Architecture (CPU vs GPU)

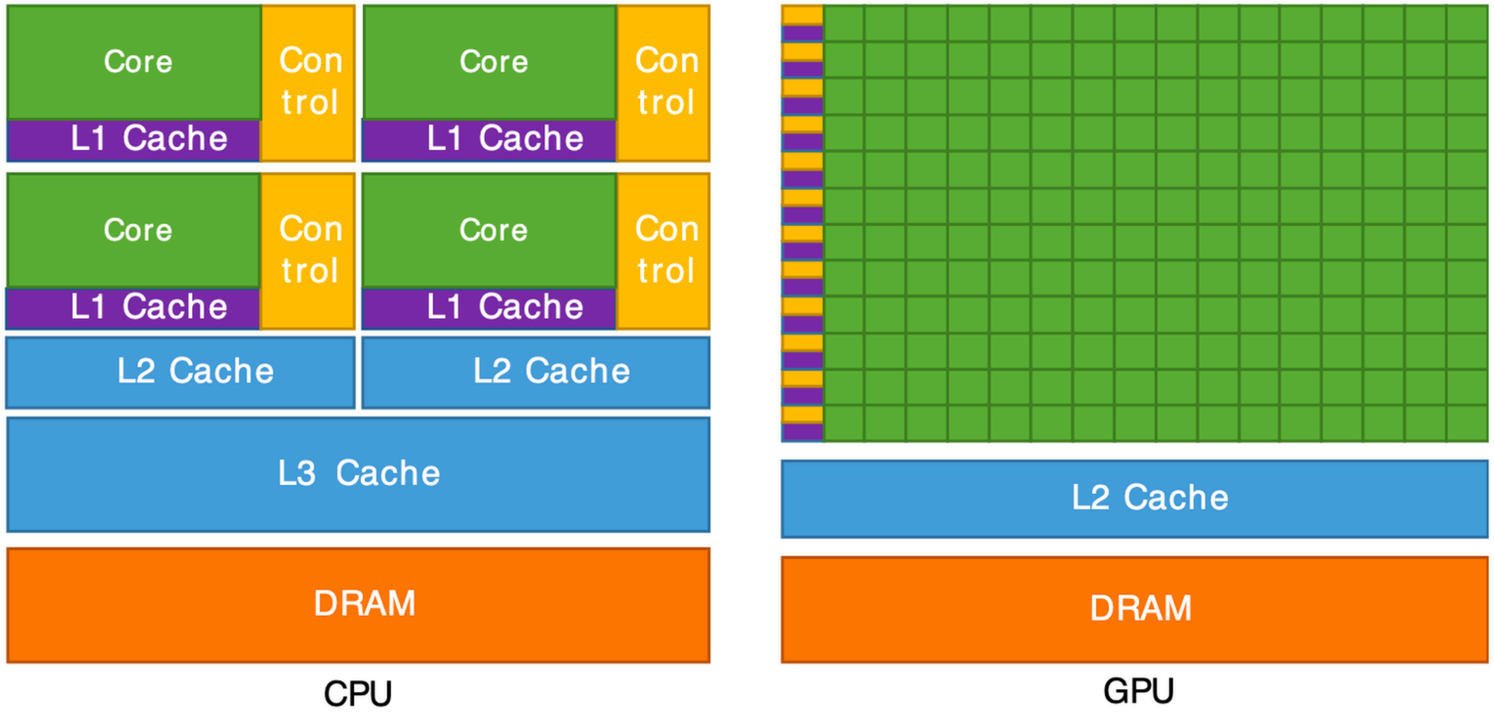
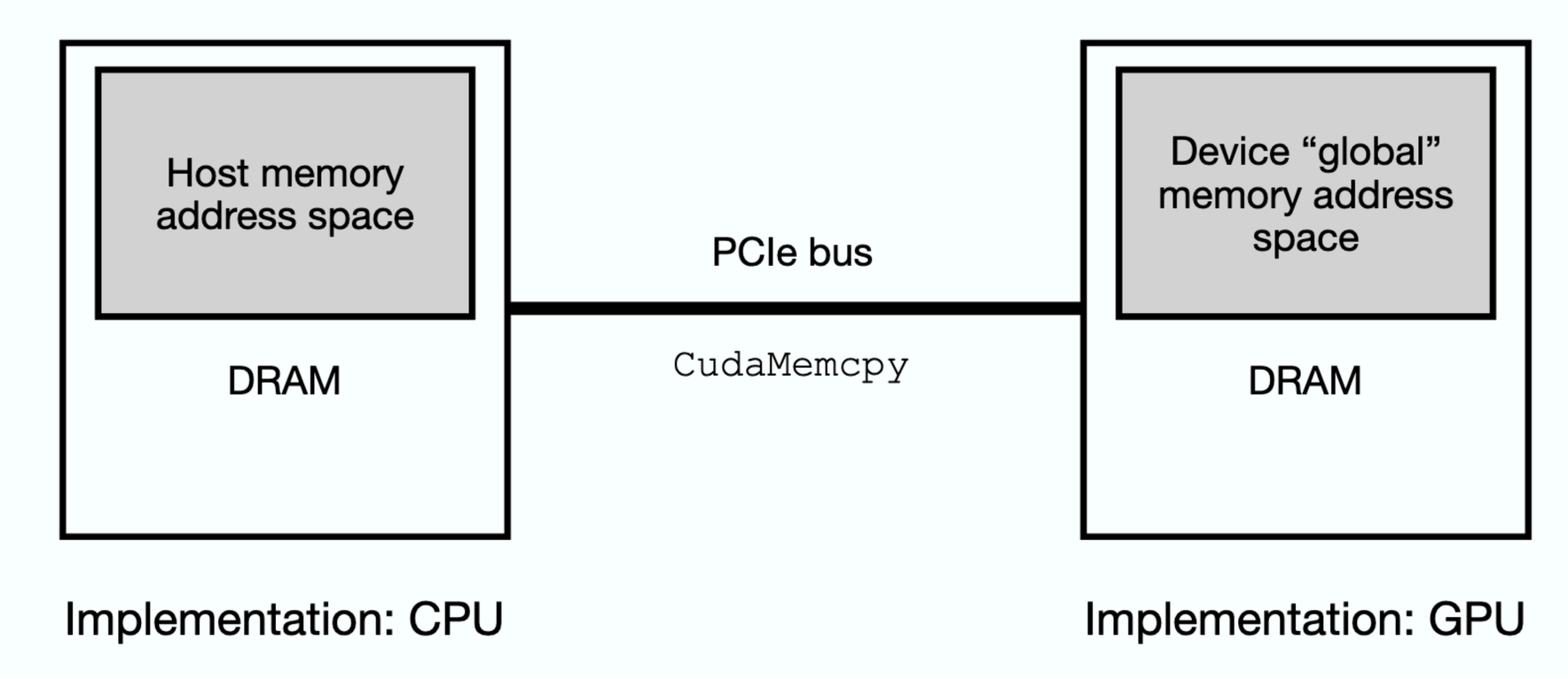


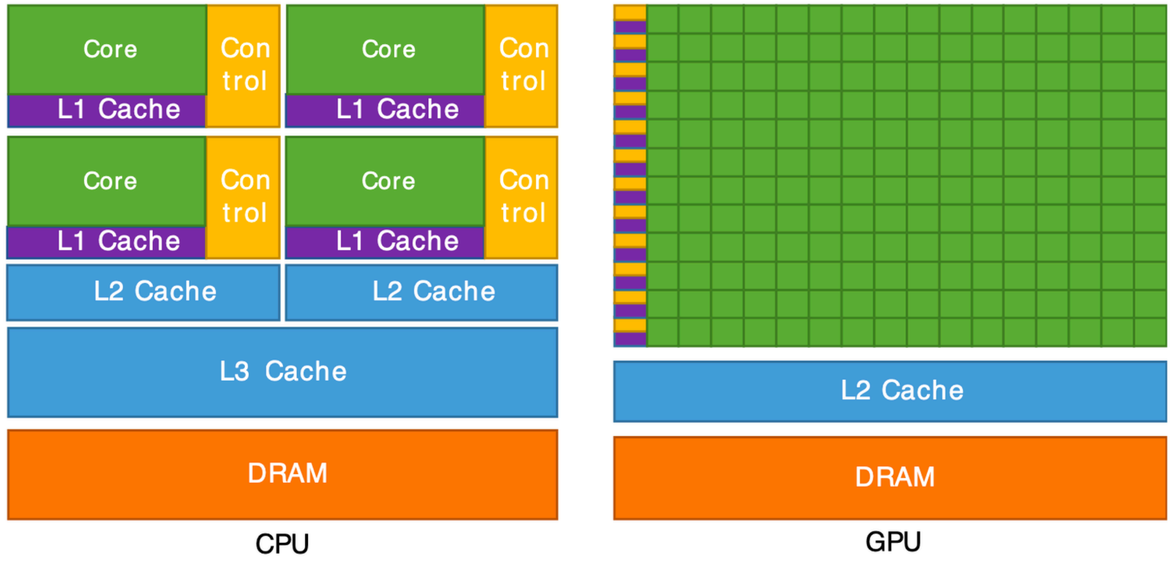
Image source: https://cvw.cac.cornell.edu/gpu-architecture/gpu-characteristics/design

Interconnection CPU & GPU



Source: https://gfxcourses.stanford.edu/cs149/fall25

Architecture (CPU vs GPU)

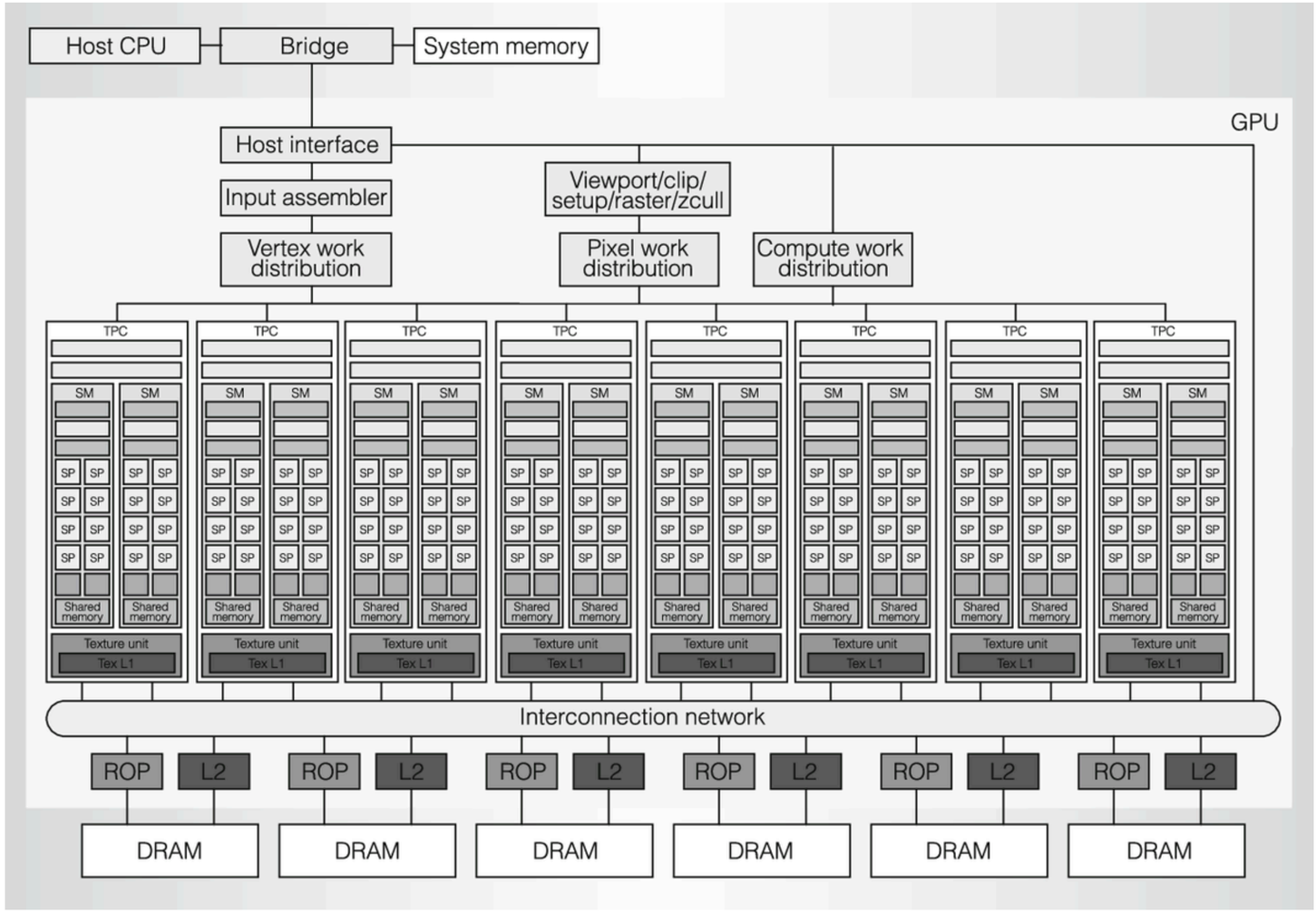


GPU >> CPU w.r.t. no. cores

GPUs have tens of thousands of threads >> No. CPU threads

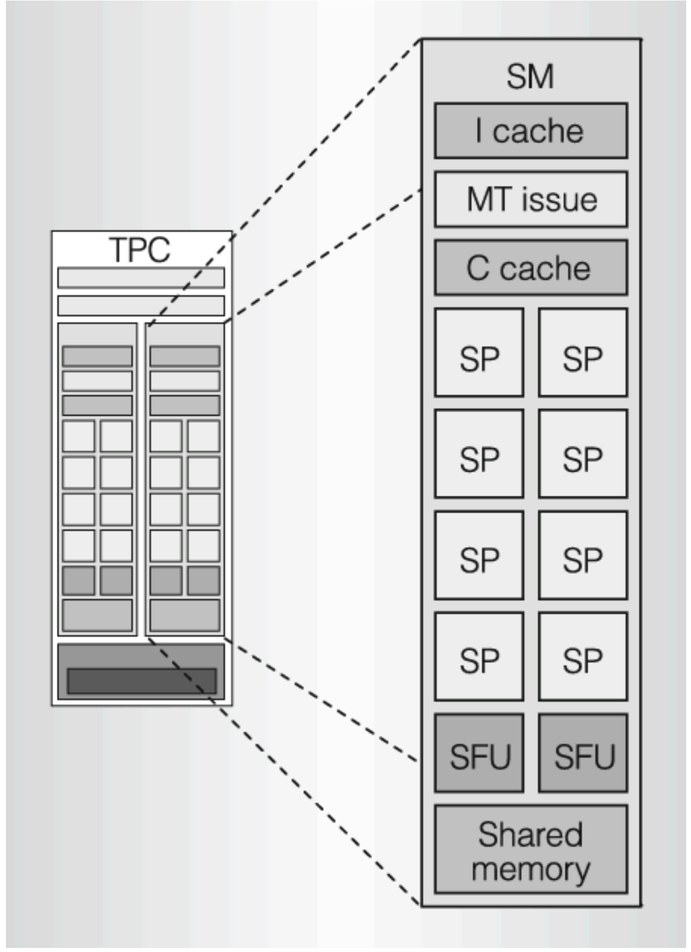
Image source: https://cvw.cac.cornell.edu/gpu-architecture/gpu-characteristics/design

Architecture GPU (Tesla, 2006)



https://www.semanticscholar.org/paper/NVIDIA-Tesla%3A-A-Unified-Graphics-and-Computing-Lindholm-Nickolls/356869aa0ae8d598e956c7f2ae884bbf5009c98c/figure/0

Architecture GPU (Tesla, 2006)



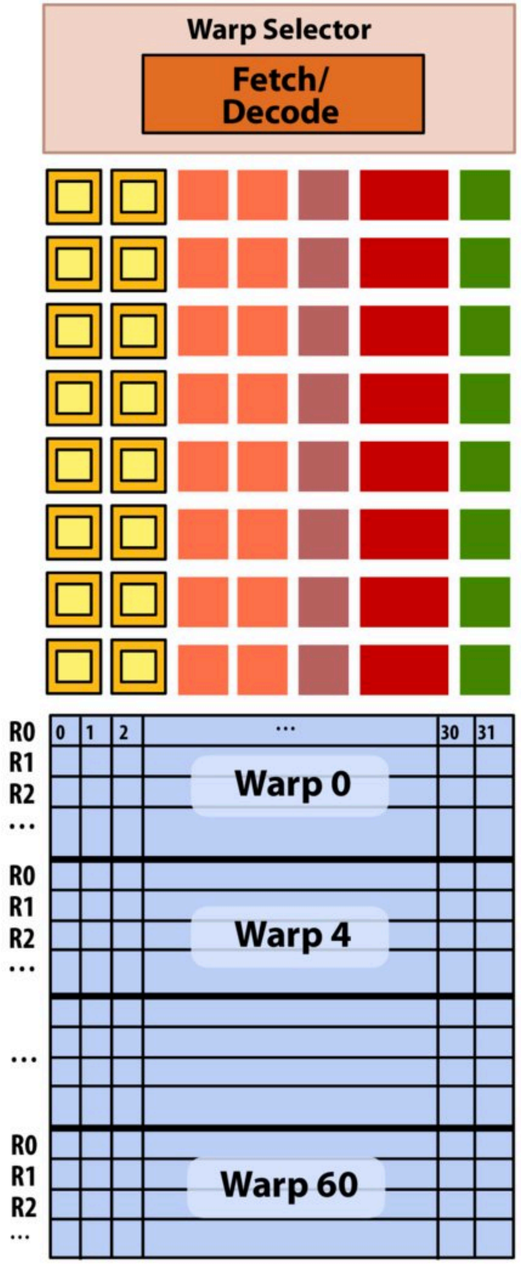
https://www.semanticscholar.org/paper/NVIDIA-Tesla%3A-A-Unified-Graphics-and-Computing-Lindholm-Nickolls/356869aa0ae8d598e956c7f2ae884bbf5009c98c/figure/4

SM of a GPU today (2026)



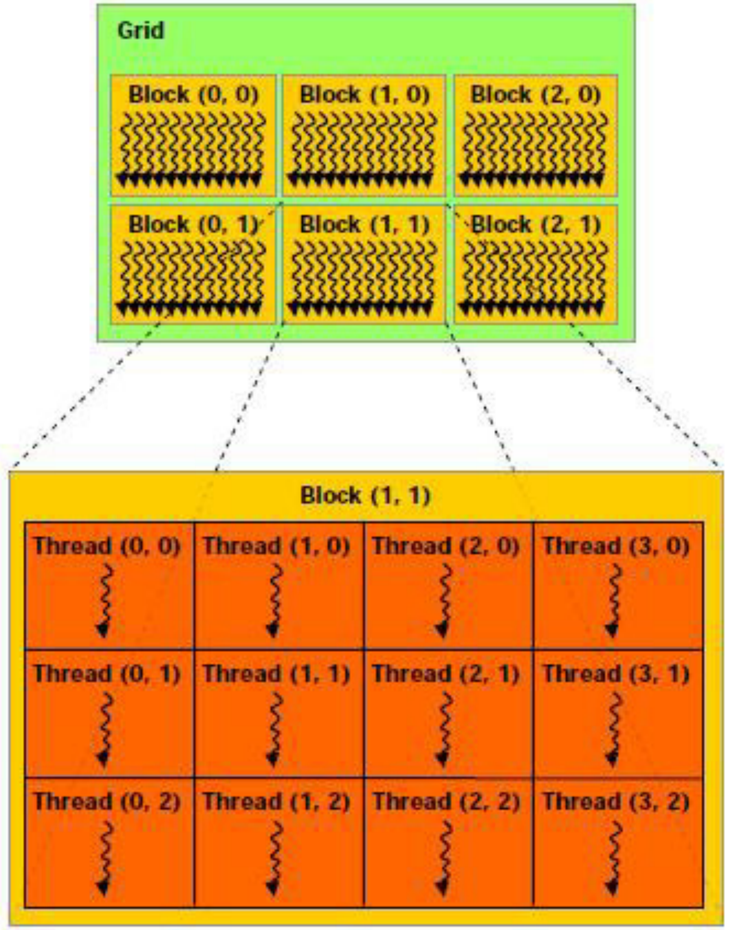
https://developer.nvidia.com/blog/inside-nvidia-blackwell-ultra-the-chip-powering-the-ai-factory-era/

Zoom-in SM



Source: https://gfxcourses.stanford.edu/cs149/fall25

GPU Programming Model



Source: https://gfxcourses.stanford.edu/cs149/fall25


NVIDIA
CUDA

?

CUDA in context

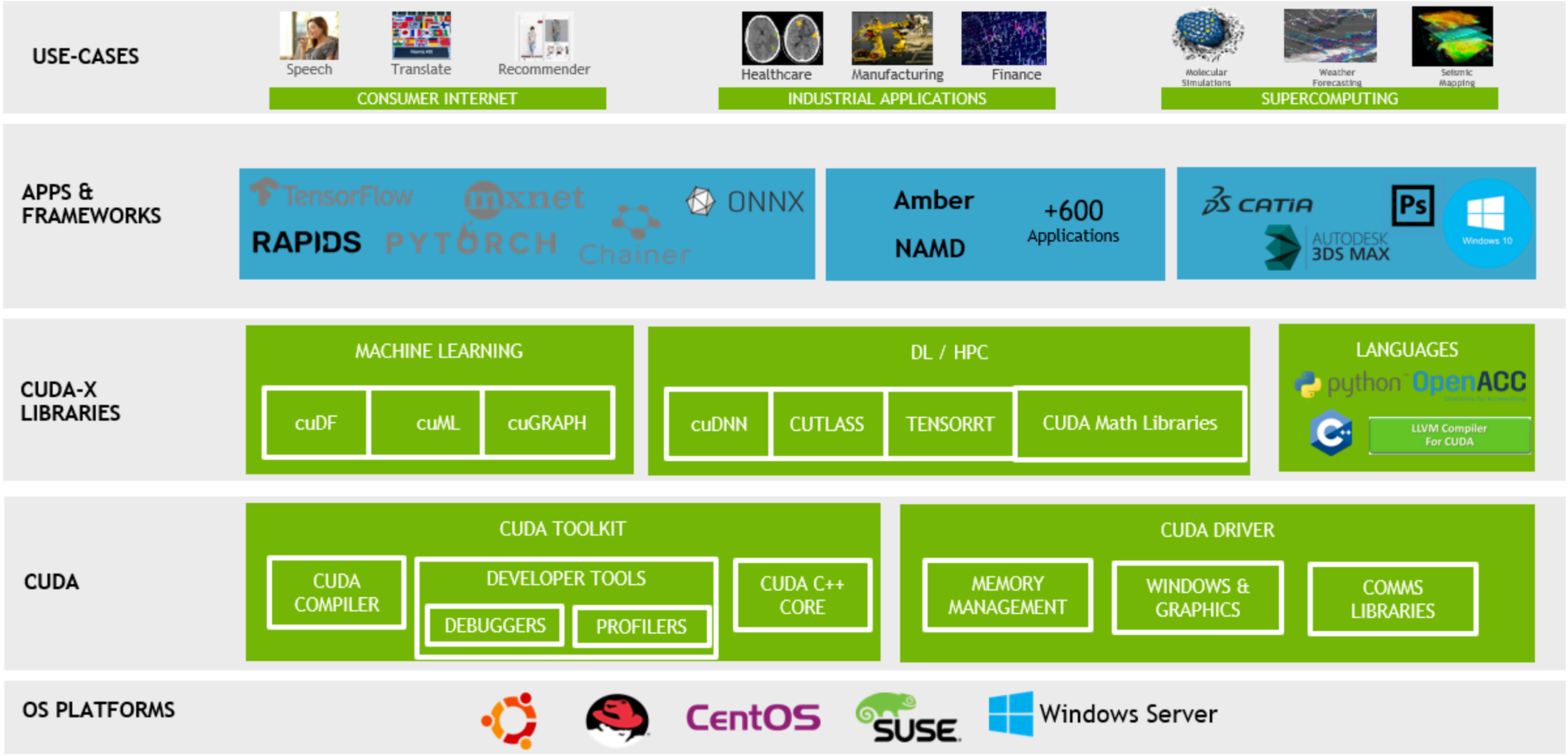


Image source: https://blogs.nvidia.com/blog/what-is-cuda-2/

What about actual execution on *HW*?

Execution on *HW*?

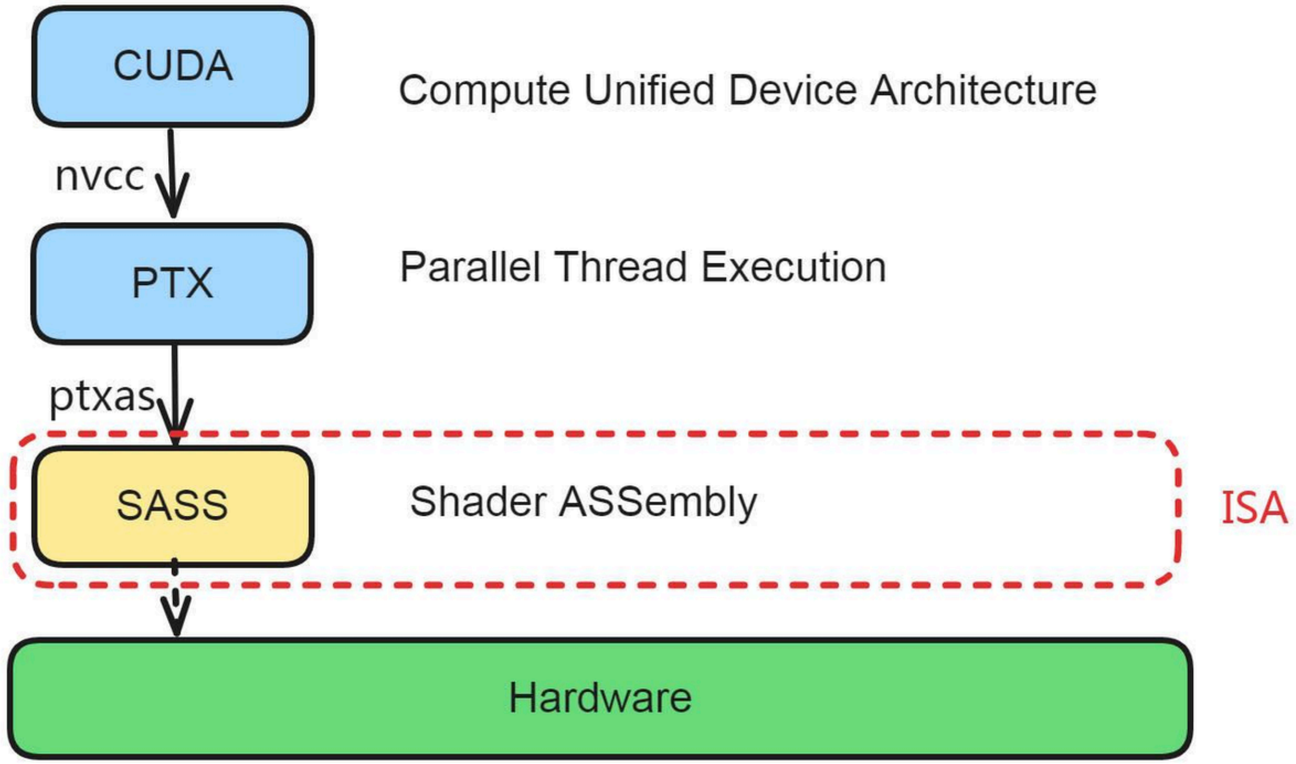


Image source: https://zhuanlan.zhihu.com/p/686198447

Hands-on exercise

Lab GPU: Architecture?